LAB REPORT

COURSE INFORMATION

- ❖ **Course Code:**
- ❖ **Course Title:**
- ❖ **Experiment No:**
- ❖ **Experiment Name:**

SUBMITTED TO

- ❖
- ❖ **Department of**
- ❖ **East West University**

SUBMITTED BY

- ❖ **Student Name :**
- ❖ **ID :**
- ❖ **Section :**
- ❖ **Department :**

East West University

Date of submission:

Table: Data for extensions and time periods for different loads

No. of obs.	Load m *Kg*	Reading of bottom end of the spring on the vertical scale l *meter*	Extension x *meter*	Total time for 30 oscillations *t / sec*	Time period T = (t / 30) *sec*	T^2 *sec^2*
1	**0 (No load)**	l_o=…………..	$(l_o - l_o)$=… ……			
2	**0.1**	l_1=…………..	$(l_1 - l_o)$=… ……			
3	**0.2**	l_2=………… …	$(l_2 - l_o)$=… ……			
4	**0.3**	l_3=…………..	$(l_3 - l_o)$=… ……			
5	**0.4**	l_4=…………..	$(l_4 - l_o)$=… ……			
6	**0.5**	l_5=…………..	$(l_5 - l_o)$=… ……			

Calculation:

$$Spring\ Constant,\ K = \frac{mg}{x} = \frac{g}{\frac{x}{m}} = \frac{9.8\ \frac{m}{s^2}}{SLOPE\ of\ Extension\ vs\ Load\ graph} = \frac{\ldots\ldots\ldots}{\ldots\ldots\ldots} = \ldots\ldots\ldots\ N/m$$

Experimental value of Effective mass (X axis ***intercept*** of ***(Time period)² Vs Load*** graph), m'

$$m' = \ldots\ldots\ldots\ldots\ \text{Kg.}$$

Mass of the spring measured by weight machine**,** $\boldsymbol{m_o}$ =…… .. ….Kg,

Standard value (Theoretical value) of Effective mass, $m' = \frac{m_o}{3} = \ldots\ldots$

$$\text{Percentage Difference} = \frac{\text{Standard value} \sim \text{Experimental value}}{\text{Standard value}} \times 100\%$$

Percentage difference = ……………

Results:

Spring constant, K =

Effective mass, m' =